

Eurostat data analysis

Geospatial data management project

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September 1, 2025



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Presentation of the datasets used

3 datasets:

- 1 Census grid 2021
- 2 Geographic accessibility
- 3 Communes



Definition of table `it_communes` and indexes for all the tables.



Calculation of the population by municipality

The formula used:

$$P_C = P_G \sum \frac{A(C \cap G)}{A(G)} \quad (1)$$

and the query performed:

```
WITH pop_calc AS (  
  SELECT  
    c."COMM_ID",  
    ROUND(SUM(ST_Area(ST_Intersection(g.geom, c.geom)) / 1000000  
      ↪ * g."T")) AS pop  
  FROM census_2021 g JOIN it_communes c ON ST_Intersects(g.geom,  
    ↪ c.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes c  
SET pop = p.pop  
FROM pop_calc p  
WHERE c."COMM_ID" = p."COMM_ID";
```



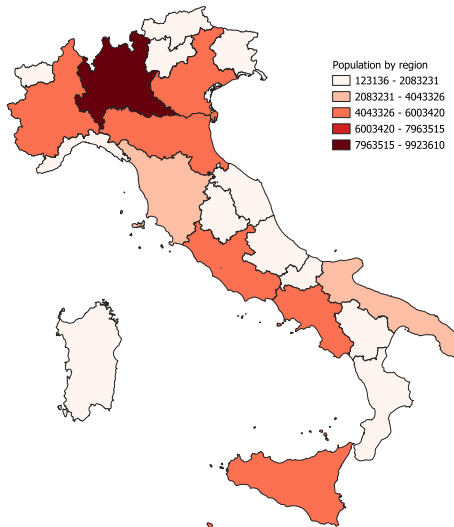
Calculation of mean accessibility value

Computation of the mean accessibility times by commune to the closest healthcare structure and to the 3 closest ones. For years 2020 and 2023.

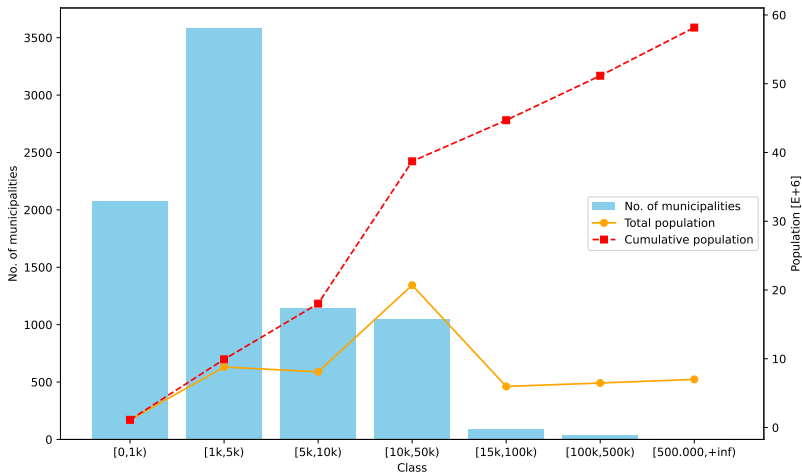
```
WITH avg_t AS (  
  SELECT  
    c."COMM_ID",  
    AVG(g.health_2020_n1) AS mean_health_2020_n1,  
    AVG(g.health_2020_n3) AS mean_health_2020_n3,  
    ...  
  FROM it_communes c JOIN grid_accessibility_health g ON  
    ↪ st_intersects(c.geom, g.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes as c  
SET  
  mean_health_2020_n1 = ROUND(t.mean_health_2020_n1::NUMERIC, 2),  
  ...  
FROM avg_t t  
WHERE t."COMM_ID"=c."COMM_ID";
```



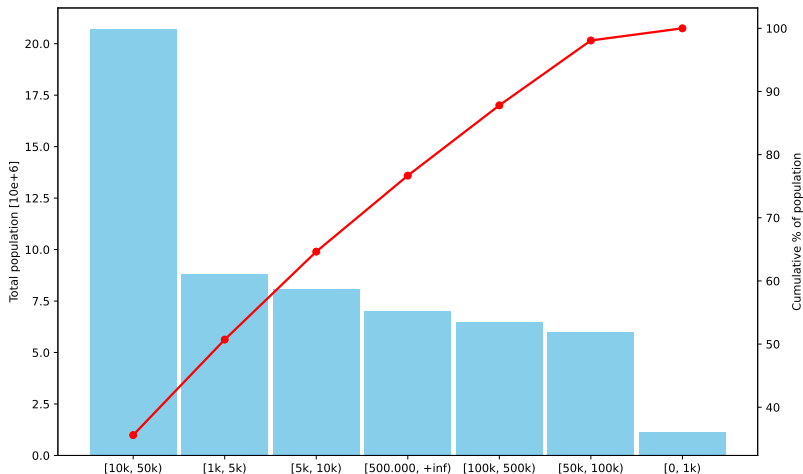
Population by region



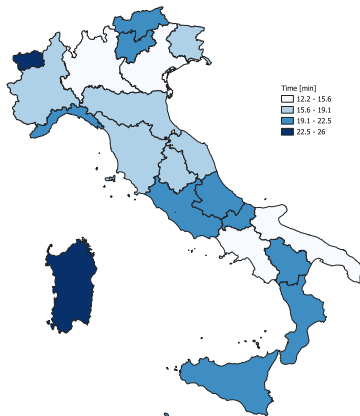
Number of communes by size



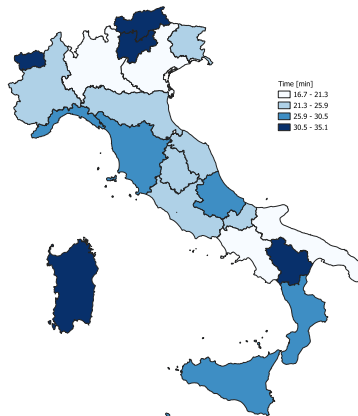
Italian population by size of municipality



Accessibility to healthcare services in Italy



(a) Acc. to the closest centre



(b) Acc. to the 3 closest centres



Categories based on travel time

No.	Travel time [min]	Accessibility
1	< 15	High
2	$[15, 30)$	Medium-high
3	$[30, 45)$	Medium-low
4	> 45	Low



Accessibility in the North of Italy

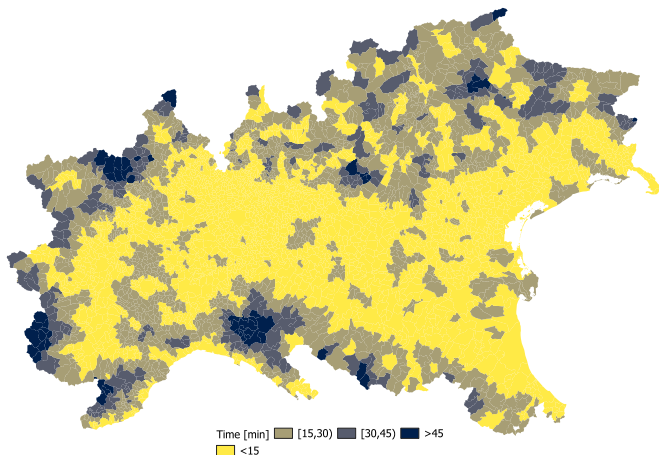


Figure: Accessibility to the closest healthcare service



Accessibility in the North of Italy

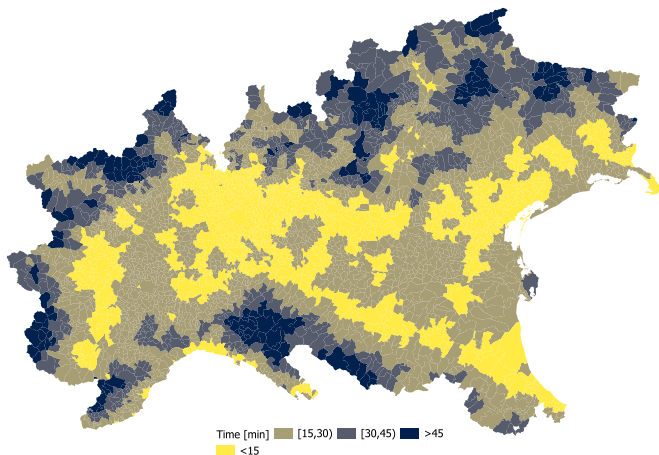


Figure: Accessibility to the 3 closest healthcare services



Distribution of municipalities by class

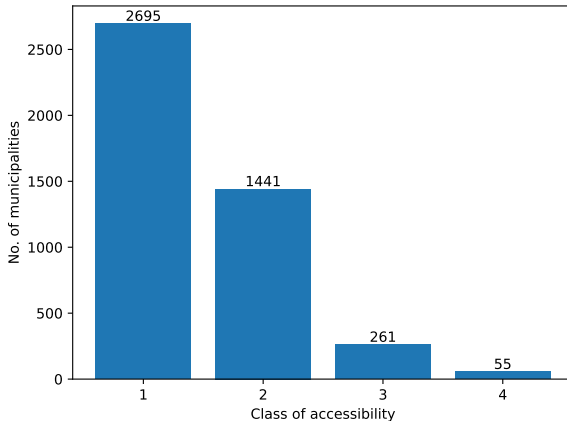


Figure: Time to reach the closest healthcare centre



Population in low accessibility areas

Class	Population	% of total
1	18 836 153	69.32
2	7 399 725	27.23
3	792 058	2.91
4	143 979	0.53
	27 171 915	



Variation between 2020 and 2023

$$\Delta_{n1}[\%] = \frac{\bar{t}_{2023n1} - \bar{t}_{2020n1}}{\bar{t}_{2020n1}} * 100$$

where

- Δ Variation as percentage
- $\bar{t}$ Mean access time in the specified year



Map of changes

